Enhancing Options Pricing: A Neural Network Approach to Volatility Forecasting

Paul Ainardi
Illinois Institute of Technology
Advisor: Professor Huiling Liao

Abstract

Volatility is the only input to the Black–Scholes formula that cannot be observed, and the one the price is most sensitive to, so a better volatility forecast ought to produce a better option price. We test a specific proposal for producing one: a sliding window whose length is set adaptively by the instantaneous energy of an Empirical Mode Decomposition (EMD) of the volatility series, feeding a masked LSTM that forecasts next-day realised volatility, whose output is then supplied to Black–Scholes. On 5,921 daily S&P 500 observations (31 January 2003 to 13 August 2026) with a strictly chronological 2,073-day out-of-sample period, the adaptive window does not improve on a fixed window of the same architecture. The direct comparison gives $p = 0.14$ on 20-day realised volatility and $p = 0.23$ on a single-day proxy, the sign of the difference changes between the two, and the two networks' forecasts correlate at 0.9999. Neither variant separates from the rule "tomorrow's volatility equals today's". On the 20-day target that rule is itself a weak benchmark: 19 of the 20 returns in tomorrow's value are already known, and simply updating the window halves its error with no fitted parameters (MSE 0.911 against 1.795), far ahead of every fitted model, including an LSTM given the daily returns themselves. On the single-day target GARCH(1,1) has the lowest error, but its edge over persistence does not survive a multiple-comparison correction once the serial dependence of the loss differential is allowed for. Re-estimating every model each year changes none of this. Passed through Black–Scholes to price a one-day ATM call, every volatility input gives a price within 8.3% MSE of every other, none significantly different from the best, against an estimated ceiling of $R^2 = 0.240$ for any model at this horizon. A small recurrent pricer does better than all of them ($R^2 = 0.137$ against 0.028), though not significantly after correction, and it gains nothing from the LSTM forecast. Diagnostics suggest why the window rule fails: its typical length is set by the number of decomposition modes rather than by the market, the regime dependence it does have points the wrong way, and about half of its day-to-day movement is an edge effect of reading a causal decomposition at its last sample.

Index Terms

Volatility forecasting, option pricing, LSTM, empirical mode decomposition, forecast evaluation, Diebold–Mariano test.

I. Introduction

In finance, an option is a contract which conveys to its owner, the holder, the right, but not the obligation, to buy or sell an underlying asset at a specified strike price on a specified date, for a European option.

Since Louis Bachelier's work in 1900, a great deal of effort has gone into finding a fair price for an option, one that, at time zero, confers no advantage to either buyer or seller. In the decades that followed, mathematical finance evolved rapidly: in 1973, Black, Scholes, and Merton introduced the celebrated risk-neutral valuation framework and closed-form formula for European options; soon after, Cox, Ross, and Rubinstein proposed the binomial tree model to approximate that valuation in a discrete setting. In the late 1970s and '80s, extensions such as Merton's jump-diffusion model and stochastic volatility formulations by Hull–White and Heston allowed more realistic dynamics of underlying assets. The 1990s saw the advent of local volatility models (Dupire, Derman–Kani) and fast numerical techniques, finite-difference schemes, Monte Carlo simulation with variance reduction, and FFT-based methods to calibrate to observed market smiles and surfaces. In the 2000s and beyond, practitioners refined tools like the SABR model for interest-rate derivatives, explored Lévy processes, and embraced machine-learning approaches to capture complex payoffs and path-dependencies [13], [14], [10], [11].

This study focuses on modern LSTM and RNN techniques [31] compared to the Black–Scholes formula [22] given by the PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

Of the formula's inputs, the price, the strike and the time to expiry are known when the option is written. Volatility is not. It has to be estimated, and it is the input the price is most sensitive to. The hypothesis we test is the natural one: if the formula is given a better estimate of tomorrow's volatility, the resulting price should be fairer. A large literature pursues exactly this, with GARCH-family models [23], [20], [2], [25], neural architectures [6], [18], [17], [7], [3], hybrid decomposition schemes [12], [1], and machine-learning pricers that learn the price directly [4], [8]. Comparisons between these families are less settled than the volume of work suggests [5], [19].

The particular idea under test is that the *window* of history a recurrent model conditions on should not be fixed, but should adapt to the prevailing volatility regime, and that an Empirical Mode Decomposition of the volatility series [30] provides a

principled way to set it. The same question, how long a sliding window should be when the signal is not stationary, comes up well outside finance [15]. A masked LSTM makes such a variable-length window trainable. The idea is appealing, and this paper tests whether it works.

Answering that question honestly turns out to depend far more on the experimental design than on the architecture, for three reasons that recur throughout this literature. First, realised volatility over a rolling window is mechanically autocorrelated, so a high R^2 can be produced by a model with no skill whatever. Naive rules must be reported or the numbers cannot be interpreted, and the usual one, persistence, is not the strongest. Second, a decomposition such as EMD is normally computed over an entire series, which quietly makes every window length a function of the future; it must be recomputed causally at each step, at considerable expense. Third, an option-pricing target assembled from quantities observable at the forecast origin turns forecasting into interpolation. We design around all three from the start, verify the design programmatically (Section VI), and report what the method does. Our contributions are: (i) a causal implementation of the EMD dynamic-window masked LSTM; (ii) a direct test of the dynamic window against a fixed window, alongside the classical benchmarks this hypothesis has to clear, including a parameter-free window update that the usual persistence benchmark leaves out, with significance tests that allow for serially dependent losses and a correction for multiple comparisons; (iii) two checks that the outcome is not an artefact of our own setup, one re-estimating every model each year and one giving the network the daily returns; and (iv) diagnostics that explain the outcome rather than merely reporting it.

II. Method

The method has two stages. Stage 1 outputs a volatility forecast for day $t+1$; stage 2 produces an option price, either by substituting that forecast into Black–Scholes or by feeding it to a recurrent pricer along with other market features.

A. Volatility targets

The feature is past realised volatility observed through a sliding window; the primary target is tomorrow's realised volatility. Realised volatility is the 20-day rolling annualised standard deviation of log-returns,

$$\sigma_t^{\mathrm{rv}} \;=\; \sqrt{\frac{252}{n-1}\sum_{i=0}^{n-1}\bigl(r_{t-i}-\bar{r}\bigr)^2}, \qquad r_t=\ln\frac{S_t}{S_{t-1}}, \quad n=20, \tag{1}$$

expressed in annualised percent. Because σ_t^{rv} is a 20-day rolling window, $\sigma_{t+1}^{\mathrm{rv}}$ and σ_t^{rv} share 19 of their 20 returns, and those 19 returns are already observed at t. This shapes the evaluation in two ways. The persistence rule $\hat{\sigma}_{t+1}=\sigma_t^{\mathrm{rv}}$ looks very hard to beat, but it throws that information away: recomputing the window from the 19 known returns and an estimate of the single unknown one is a stronger naive benchmark, and we report both. And forecast errors are serially dependent, which the inference in Section III has to allow for.

We therefore also report the single-day proxy

$$\sigma_{t+1}^{\mathrm{rv},(1)} \;=\; |r_{t+1}|\sqrt{\pi/2}\,\sqrt{252}\times 100, \tag{2}$$

which is unbiased for σ_{t+1} under conditional normality (for $r\sim\mathcal{N}(0,\sigma^2)$, $\mathbb{E}|r|=\sigma\sqrt{2/\pi}$) and is far less autocorrelated (Fig. 5). It is a much noisier measurement of the same underlying quantity, and the genuinely difficult version of the question.

For the pricing stage, the target is the ex-post fair value of a one-day ATM call struck at S_t,

$$V_t^{*} \;=\; \tfrac{1}{2}\,\bigl|S_{t+1}-S_t\bigr|. \tag{3}$$

Under a zero-drift symmetric one-day return distribution, $\mathbb{E}_t[\max(S_{t+1}-S_t,0)]=\frac{1}{2}\mathbb{E}_t|S_{t+1}-S_t|$, so V_t^{*} is an unbiased, if very noisy, realisation of the quantity Black–Scholes estimates, and it is not observable at t.

B. EMD-based dynamic window

To capture non-stationary cycles we apply Empirical Mode Decomposition [30] to the volatility series X_t, extracting Intrinsic Mode Functions $\{\mathrm{IMF}_k(t)\}_{k=1}^{K}$ by the standard sifting procedure: locate the local extrema, fit upper and lower envelope splines, subtract their mean, and repeat until the result satisfies the IMF conditions, then repeat on the residual. We compute the instantaneous amplitude via the Hilbert transform $\mathcal{H}[\cdot]$ and, for each mode, a normalised power

$$\pi_k(t) \;=\; \frac{\bigl|\mathcal{H}[\mathrm{IMF}_k](t)\bigr|}{\sqrt{\dfrac{1}{|\mathcal{W}_t|}\sum_{u\in\mathcal{W}_t}\bigl|\mathcal{H}[\mathrm{IMF}_k](u)\bigr|^2}}, \qquad \pi(t)=\sum_{k=1}^{K}\pi_k(t), \tag{4}$$

and set the window length

$$\tau_t \;=\; \min\bigl(W_{\max},\,\max(1,\,\mathrm{round}(\pi(t)))\bigr). \tag{5}$$

$\mathcal{W}_t$ is a *trailing* window, $\mathcal{W}_t = \{t - L + 1, \ldots, t\}$ with $L = 500$ trading days, and the decomposition is recomputed at every t from that window alone. It costs one decomposition per trading day instead of one for the whole sample, and it is not optional. Under a single whole-sample decomposition the envelope splines at time t are fitted through extrema that have not yet occurred, so τ_t becomes a function of the future and every downstream forecast is contaminated. We verify causality directly rather than by inspection: corrupting the series after any index t_0 leaves $\pi(t)$ for $t < t_0$ bitwise unchanged (Section VI). Causality has a price, though. The rule has to read $\pi(t)$ at the last sample of each window, which is exactly where the envelope splines and the FFT-based Hilbert transform are least reliable. Section V measures how much this matters.

C. Masked LSTM

For each forecast origin $t \geq W_{\max} - 1$ we form an input matrix $\mathbf{X}_t \in \mathbb{R}^{W_{\max} \times 2}$ whose first column holds the most recent τ_t standardised volatility observations, zero-padded on the left, and whose second column is a binary mask marking which rows are real data:

$$[\mathbf{X}_t]_{i,1} = \begin{cases} X_{t-\tau_t+i}, & i > W_{\max} - \tau_t, \\ 0, & \text{otherwise}, \end{cases} \qquad [\mathbf{X}_t]_{i,2} = m_i \in \{0, 1\}.$$

Every entry uses information available at t inclusive, and the target is a function of $t + 1$ only. The mask is required because a padded zero is a legitimate value in standardised units and would otherwise be indistinguishable from data.

This matrix is fed to a two-layer masked LSTM. Writing $(h_{i-1}^{(\ell)}, c_{i-1}^{(\ell)})$ for the hidden and cell states of layer $\ell \in \{1, 2\}$ before processing row i, the mask is applied to the incoming states,

$$\bar{h}_{i-1}^{(\ell)} = m_i \, h_{i-1}^{(\ell)}, \qquad \bar{c}_{i-1}^{(\ell)} = m_i \, c_{i-1}^{(\ell)},$$

and the cell then updates exactly as a standard LSTM, using $\bar{h}$ and $\bar{c}$ in place of the raw previous states:

$$i_t = \sigma(W_i x_t + U_i \bar{h}_{t-1} + b_i), \quad f_t = \sigma(W_f x_t + U_f \bar{h}_{t-1} + b_f), \quad o_t = \sigma(W_o x_t + U_o \bar{h}_{t-1} + b_o),$$
$$g_t = \tanh(W_c x_t + U_c \bar{h}_{t-1} + b_c), \qquad c_t = f_t \odot \bar{c}_{t-1} + i_t \odot g_t, \qquad h_t = o_t \odot \tanh(c_t).$$

Finally we project the top-layer hidden state at the last row to a scalar, $\hat{\sigma}_{t+1} = W_y \, h_{W_{\max}}^{(2)} + b_y$, and train by minimising mean squared error with Adam, gradient-norm clipping at 5, and a plateau learning-rate schedule.

Two implementation details materially affect the results and are stated because omitting them would make the numbers irreproducible. First, the *target* is standardised using training-slice location and scale, and the transform is inverted at prediction time. The output head is a linear map of a $\tanh$-bounded state, so regressing raw annualised percent (training mean near 16, peaks near 100 in the test period) requires large output weights, optimises poorly, and makes the network underfit the upper tail badly. This is a property of the parameterisation, not of the architecture, and standardising removes it; forecasts are floored at zero, since a volatility cannot be negative. Second, the mask zeroes the incoming state at a padded row, but the cell is still evaluated there, so the state entering the first real row is the bias-driven output of the last padding step rather than exactly zero. The equations above describe exact freezing and the implementation delivers an approximation to it; we measure the residual at 4×10^{-3} (Section VI) and note it as a known limitation rather than claiming exactness.

Both variants above see only σ^{rv}, which is already a 20-day average. A network fed that series cannot tell which daily return is about to leave the window, so we also train a third variant, the ***LSTM on returns***: the same two-layer network over a fixed window of 20 or 30 days whose rows hold both σ^{rv} and the absolute daily log return $|r|$, each standardised on the training slice. It is not part of the window comparison; it tests whether the LSTM's showing is limited by what it is given.

D. Pricing stage

To obtain a price we either substitute $\hat{\sigma}_{t+1}$ into the ATM Black–Scholes formula with $T = 1/252$, or feed it to a two-layer GRU together with other market features (log return, σ_t^{rv}, VIX, the 13-week T-bill yield). The recurrent pricer is a standard architecture, is not claimed as a contribution, and uses the same target-standardisation treatment described above. One more detail matters here. V^* is measured in index points and scales with S_t, while none of the GRU's inputs is a price level, and the index stands 1.75 times higher in the pricing test period than in the period the GRU is trained on. Asked for index points directly, the network would have to extrapolate a level it never saw. It is therefore trained on V_t^*/S_t, and its output is multiplied by S_t.

III. Experimental design

Data. Daily S&P 500 OHLCV, the CBOE VIX index, and the 13-week Treasury bill yield, from Yahoo Finance, 2 January 2003 to 14 August 2026. After computing rolling estimators and dropping warm-up rows the modelling sample is 5,921 observations, 31 January 2003 to 13 August 2026. Features are price-derived only: OHLCV plus realised-volatility estimators (Parkinson [27], Garman–Klass [28], Yang–Zhang [29], EWMA [26]). We deliberately exclude macroeconomic series such as CPI, GDP and unemployment: they are published with release lags that are easy to align incorrectly, and a misalignment there

is indistinguishable from forecasting skill. The risk-free input is the 13-week T-bill yield, matched to the one-day horizon of the option being priced.

Splits. Strictly chronological, never shuffled, and nested so the pricing stage never consumes a forecast that was in-sample for stage 1:

- *Stage 1* trains on the first 55% of observations (through 7 January 2016), uses the next 10% for early stopping and the learning-rate schedule (through 15 May 2018), and forecasts over the remaining 35%: 2,073 trading days from 15 May 2018 to 13 August 2026, a period that contains the COVID crash and the 2022 drawdown.
- *Stage 2* operates only inside that region, splitting it $60/10/30$, so its 622-day test set (21 February 2024 to 13 August 2026) sees only volatility forecasts that were out-of-sample for stage 1.

All scalers, OLS coefficients, and GARCH parameters are estimated on the relevant training slice only and applied forward; nothing is refit on a validation or test window. In the main tables every model, classical or neural, is therefore estimated once and held fixed for eight years of forecasting. That is the same handicap for all of them, but it is not how they would be used in practice, so we repeat stage 1 with yearly re-estimation. The test period is cut into 9 consecutive blocks of 252 trading days. Before each block, GARCH(1,1) and the HAR-type regression are refitted on all data up to that date, and every network is retrained on all of it except the most recent 592 days, which play the role of the validation slice exactly as in the static design (so the first block reproduces it). Hyperparameters stay at their static selection, and parameter-free methods are unchanged.

Hyperparameters. Hidden size $\in \{32, 64\}$, dropout $\in \{0.1, 0.2\}$, and $W_{\max} \in \{10, 20, 30\}$ are selected by grid search on validation MSE, run *separately for each combination of target and window scheme*. Searching once and reusing the result would favour whichever variant the search was run on, which is precisely the comparison this paper is about. The LSTM on returns is searched the same way with its window in $\{20, 30\}$, long enough to contain the return leaving the 20-day window; validation chose 30 days on the 20-day target and 30 on the single-day one. Selected configurations are recorded in `results/hyperparams.json`.

Benchmarks. Persistence ($\hat{\sigma}_{t+1} = \sigma_t^{\mathrm{rv}}$); a random walk on the single-day proxy ($\hat{\sigma}_{t+1} = \sigma_t^{\mathrm{rv},(1)}$); RiskMetrics EWMA with $\lambda = 0.94$; GARCH(1,1) [23] by Gaussian QMLE on the training slice; a HAR-type regression of the target on σ_t^{rv} and its 5- and 22-day averages, fit by OLS on the training slice; the VIX index; and a 20-day Yang–Zhang estimator. The HAR-type regression borrows Corsi's structure [24] but not his inputs: HAR-RV proper builds its three terms from daily realised variance, which on the 20-day target is the weaker of the two versions, so we keep the stronger one. On the 20-day target we add the *window update*, which keeps the 19 returns of $\sigma_{t+1}^{\mathrm{rv}}$ that are already known at t, replaces the square of the unknown return by its EWMA expectation, and recomputes the sample standard deviation. It has no fitted parameters. For pricing we add a constant predictor at the training mean, Black–Scholes with a constant σ equal to the training mean of σ^{rv}, and a persistence predictor V_{t-1}^{*}. The constant-σ row is the honest "no volatility forecast" benchmark, since it scales with S_t like every other Black–Scholes row; the constant in index points does not.

Inference. Every neural forecast is the average of five networks trained with seeds $\{42, 0, 1, 7, 2024\}$. The tables report the MSE of that average, with the standard deviation of the five single-seed MSEs beside it. Differences are tested with a Diebold–Mariano statistic [32] on squared-error differentials, with Newey–West standard errors [33]. The loss differentials are serially dependent even though every forecast is one step ahead. Volatility clustering makes squared errors persistent (first-order autocorrelation 0.25 for the HAR-type regression against persistence), and on the 20-day target the return that leaves the window comes back in the persistence error twenty days later (autocorrelation -0.39 at lag 20). The rule-of-thumb bandwidth would use only 7 lags here, which understates the long-run variance and gives p-values that are too small. Every comparison in the paper therefore uses at least 19 lags, and Table IV repeats the comparisons we interpret with 40 lags and with a circular block bootstrap [34]. Because each table compares many methods against one benchmark, we report Holm–Bonferroni adjusted p-values [35] alongside the raw ones and interpret only the adjusted column. Alongside MSE and MAE we report QLIKE, which is robust to noise in a volatility proxy [21] provided the proxy is unbiased for the variance. On the single-day target that means the squared return, $\frac{2}{\pi}(\sigma_{t+1}^{\mathrm{rv},(1)})^2$, because $(\sigma_{t+1}^{\mathrm{rv},(1)})^2$ itself overstates the variance by a factor $\pi/2$.

IV. Results

A. Does the dynamic window beat a fixed window?

This is the paper's central question, it is a single pre-specified comparison between two configurations of one architecture, and it is therefore tested directly rather than inferred from separate comparisons against a third method. Table I gives the result.

TABLE I

THE CENTRAL COMPARISON: THE EMD DYNAMIC WINDOW AGAINST A FIXED WINDOW OF THE SAME ARCHITECTURE, EACH AT ITS OWN BEST HYPERPARAMETERS. ENTRIES ARE TEST MSE OF THE FIVE-SEED ENSEMBLE FORECAST, ± THE S.D. OF THE SINGLE-SEED MSES. DM IS THE DIEBOLD–MARIANO STATISTIC FOR DYNAMIC VERSUS FIXED (NEWEY–WEST, BANDWIDTH AT LEAST 19); A POSITIVE VALUE MEANS THE DYNAMIC WINDOW IS WORSE.

Target	EMD dynamic	Fixed	DM	p	Matched $W_{\max}$
$\sigma^{\mathrm{rv}}_{t+1}$ (20-day rolling)	1.929 ± 0.035	1.861 ± 0.034	+1.49	0.137	yes
$\sigma^{\mathrm{rv},(1)}_{t+1}$ (single-day)	266.66 ± 1.73	268.60 ± 1.12	−1.20	0.230	no (10 vs 20)

The answer is no on both targets. On the 20-day target the dynamic window is slightly worse than the fixed window, on the single-day target slightly better, and neither difference is significant. Table IV gives the same answer under every bandwidth we tried except the shortest, under which the dynamic window is significantly *worse* on the 20-day target ($p = 0.045$). The single-day comparison is also confounded, since validation chose $W_{\max} = 10$ for the dynamic window and 20 for the fixed one, so it compares two window lengths as well as two window rules. On the 20-day target, where both use $W_{\max} = 10$, the ordering is stable across seeds: the dispersion (± 0.034 fixed, ± 0.035 dynamic) is about half the gap between the two variants, so the dynamic window is reproducibly a little worse there rather than randomly so. Reproducible is not the same as consequential, though. Relative to the sampling variability of the loss differential over 2,073 days, a gap that size cannot be told apart from zero, and over the test period the two networks' forecasts correlate at 0.9999. The adaptive window barely changes the forecast, and what it changes goes the wrong way on the target it was designed for.

B. Stage 1: forecasting realised volatility

Table II places the three LSTM variants against the classical benchmarks on next-day $\sigma^{\mathrm{rv}}_{t+1}$.

TABLE II

NEXT-DAY 20-DAY REALISED VOLATILITY $\sigma^{\mathrm{rv}}_{t+1}$, TEST SET (2,073 DAYS). NEURAL ENTRIES ARE THE MSE OF THE FIVE-SEED ENSEMBLE FORECAST; ± GIVES THE S.D. OF THE FIVE SINGLE-SEED MSES. DM IS THE DIEBOLD–MARIANO STATISTIC AGAINST PERSISTENCE (NEWEY–WEST, BANDWIDTH AT LEAST 19); POSITIVE MEANS WORSE THAN PERSISTENCE. p_{Holm} CONTROLS THE FAMILY-WISE ERROR RATE ACROSS THIS TABLE.

Method	MSE	MAE	R^2	QLIKE	DM	p	p_{Holm}
Window update (19 of 20 returns known)	**0.911**	0.502	0.992	0.0082	−3.44	<0.001	0.005
HAR-type regression on σ^{rv}	1.644	0.637	0.986	0.0121	−2.13	0.033	0.133
Persistence ($\hat{\sigma}_{t+1} = \sigma^{\mathrm{rv}}_t$)	1.795	0.631	0.985	0.0124	–	–	–
Masked LSTM, fixed window	1.861 ± 0.034	0.683	0.984	0.0121	+0.46	0.644	0.908
Masked LSTM, EMD dynamic window	1.929 ± 0.035	0.688	0.984	0.0122	+0.75	0.454	0.908
LSTM on returns, fixed window	3.188 ± 0.342	0.792	0.973	0.0119	+1.14	0.256	0.767
GARCH(1,1), QMLE	8.678	1.969	0.926	0.0511	+2.85	0.004	0.026
EWMA ($\lambda = 0.94$)	10.893	2.057	0.907	0.0521	+3.34	<0.001	0.006
Yang–Zhang (20d)	15.018	2.342	0.872	0.0635	+2.41	0.016	0.080
VIX (30-day implied)	52.929	5.519	0.549	0.2533	+4.62	<0.001	<0.001

Lower MSE, MAE and QLIKE are better; higher R^2 is better.

Four observations. First, the best forecast in the table fits nothing. The window update halves the error of persistence (MSE 0.911 against 1.795, $p_{\mathrm{Holm}} = 0.005$), and it stays significant under every alternative in Table IV. Persistence looks hard to beat on this target only because it ignores the 19 returns of tomorrow's window that are already known. Any claim of forecasting skill on $\sigma^{\mathrm{rv}}_{t+1}$ should be measured against the window update, not against persistence.

Second, among the methods that do fit something, none significantly beats persistence once multiplicity is accounted for. The HAR-type regression comes closest (MSE 1.644 against 1.795), but its raw p-value of 0.033 becomes 0.133 after the Holm correction, the block bootstrap gives 0.069 even before correction, and we do not claim it. Corsi's original construction from daily components does much worse on this target (MSE 5.09), because a daily proxy says little about a 20-day statistic.

Third, the two LSTMs fed σ^{rv} land close to persistence and are statistically indistinguishable from it. They clearly beat GARCH(1,1), EWMA, Yang–Zhang and VIX on this target, which at least shows the network trains properly. It just has nothing to add over the naive rule. Section IV-D comes back to the fact that this ordering reverses at the pricing stage.

Fourth, giving the network the daily returns does not close the gap to the window update. The LSTM on returns has every input the update uses, and its MSE is 3.188, worse than persistence. Most of that is one episode: 65% of its squared error falls in the 70 trading days around the COVID crash (20 February 2020 to 31 May 2020), where standardised absolute returns reached values far outside anything in the training slice and the network extrapolated badly. Outside that window it does beat persistence and the fixed-window LSTM (MSE 1.154 against 1.292 and 1.264), so it does learn something from the returns. But it is still far from the window update there (0.686), which does the same bookkeeping exactly and without parameters. On 3,256 training days the network does not discover a rule that is one line of arithmetic.

Fifth, the clustering of R^2 near 0.98 in every row is a property of the target's autocorrelation, not evidence of skill. That is why the naive rows have to be in the table for it to be read at all.

TABLE III
NEXT-DAY SINGLE-DAY VOLATILITY PROXY $\sigma_{t+1}^{\mathrm{rv},(1)}$, SAME TEST SET (2,073 DAYS) AND PROTOCOL AS TABLE II. QLIKE USES $\frac{2}{\pi}(\sigma_{t+1}^{\mathrm{rv},(1)})^2$, THE SQUARED RETURN, AS THE VARIANCE PROXY.

Method	MSE	MAE	R^2	QLIKE	DM	p	p_{Holm}
GARCH(1,1), QMLE	**249.22**	10.548	0.260	1.5661	-2.20	0.028	0.221
LSTM on returns, fixed window	251.10 $\pm$ 2.20	10.285	0.255	1.5993	-2.16	0.030	0.221
VIX (30-day implied)	257.18	11.436	0.237	1.6043	-1.21	0.227	1.000
HAR-type regression on σ^{rv}	263.64	10.572	0.217	1.6354	-1.64	0.101	0.609
Masked LSTM, EMD dynamic window	266.66 $\pm$ 1.73	10.466	0.208	1.6426	-0.75	0.451	1.000
EWMA ($\lambda = 0.94$)	266.81	10.845	0.208	1.6265	-0.76	0.449	1.000
Yang–Zhang (20d)	268.06	10.340	0.204	1.6443	-0.30	0.767	1.000
Masked LSTM, fixed window	268.60 $\pm$ 1.12	10.482	0.203	1.6578	-0.46	0.644	1.000
Persistence ($\hat{\sigma}_{t+1} = \sigma_t^{\mathrm{rv}}$)	272.02	10.768	0.192	1.6827	–	–	–
Random walk ($\sigma_t^{\mathrm{rv},(1)}$)	430.35	13.829	-0.278	11010.0412	$+6.16$	<0.001	<0.001

The random walk uses yesterday's single-day proxy; persistence uses the smoother 20-day estimate. The random walk's QLIKE is dominated by days on which its forecast is close to zero.

On the single-day proxy (Table III) the ordering changes, as it should. This target is not mechanically autocorrelated, so persistence is far weaker and the spread between methods means more. GARCH(1,1) has the lowest error (MSE 249.2 against 272.0 for persistence), in line with the long-standing finding that it is hard to beat [25]. It does not clear the multiple-comparison bar, though ($p_{\mathrm{Holm}} = 0.221$), and Table IV shows how much the evidence depends on the bandwidth: with the rule-of-thumb 7 lags its raw p-value is 0.006, with the floor of 19 it is 0.028, and with 40 lags or the block bootstrap it is 0.068 or 0.055. We read it as suggestive and no more. Here the returns help the network: the LSTM on returns comes second, just behind GARCH (MSE 251.1), which makes sense since GARCH too is driven by recent squared returns. It is not significant after correction either ($p_{\mathrm{Holm}} = 0.221$). The two LSTMs fed σ^{rv} do not benefit: the dynamic-window one lands next to EWMA and the fixed-window one just above persistence, neither significantly ahead of it. The random walk on the single-day proxy is far worse than everything else (MSE 430.4), which confirms that a smoothed volatility estimate, not yesterday's noisy realisation, is the right thing to condition on.

TABLE IV
TWO-SIDED p-VALUES FOR THE COMPARISONS THE TEXT INTERPRETS, UNDER THREE NEWEY–WEST BANDWIDTHS AND A CIRCULAR BLOCK BOOTSTRAP (60-DAY BLOCKS). THE FIRST IS THE RULE-OF-THUMB BANDWIDTH (7 LAGS ON THE VOLATILITY TARGETS, 6 ON V^*); THE TABLES ABOVE USE THE SECOND. RATIO IS THE MSE OF THE FIRST METHOD OVER THE SECOND; BELOW ONE FAVOURS THE FIRST. RAW p-VALUES, BEFORE THE HOLM CORRECTION.

Comparison	Ratio	NW rule	NW 19	NW 40	Bootstrap
σ^{rv}: window update vs. persistence	0.51	<0.001	<0.001	0.002	0.006
σ^{rv}: HAR-type vs. persistence	0.92	0.008	0.033	0.091	0.069
σ^{rv}: EMD vs. fixed window	1.04	0.045	0.137	0.187	0.130
σ^{rv}: LSTM on returns vs. window update	3.50	0.025	0.096	0.152	0.085
$\sigma^{\mathrm{rv},(1)}$: GARCH(1,1) vs. persistence	0.92	0.006	0.028	0.068	0.055
$\sigma^{\mathrm{rv},(1)}$: EMD vs. fixed window	0.99	0.304	0.230	0.206	0.200
V^*: BS (GARCH(1,1)) vs. BS (constant σ)	0.83	<0.001	<0.001	<0.001	<0.001
V^*: BS (GARCH(1,1)) vs. constant	0.92	0.311	0.317	0.295	0.258
V^*: RNN (base) vs. BS (GARCH(1,1))	0.89	0.004	0.013	0.032	0.037
V^*: RNN (base) vs. rescaled BS (GARCH(1,1))	0.89	0.005	0.014	0.034	0.040

Figure 1 shows both volatility tables on a common scale, and Figure 2 shows out-of-sample predictions against realised values for the HAR-type regression and the EMD-window LSTM.

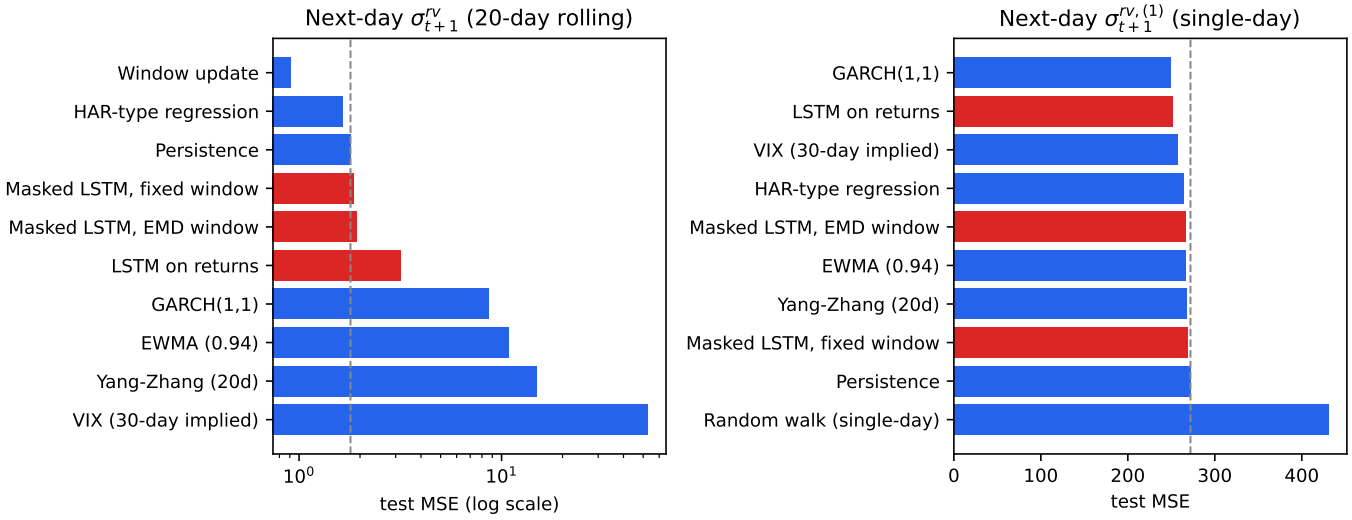


Fig. 1. Stage-1 test MSE. The dashed line marks persistence. On the 20-day rolling target (left, log scale) the window update is far ahead, the fitted models fed σ^{rv} cluster near persistence, the LSTM on returns trails them because of the 2020 crash, and the single-day variance models trail further; on the single-day target (right) GARCH(1,1) and the LSTM on returns lead, and every method except the random walk is within about a tenth of persistence.

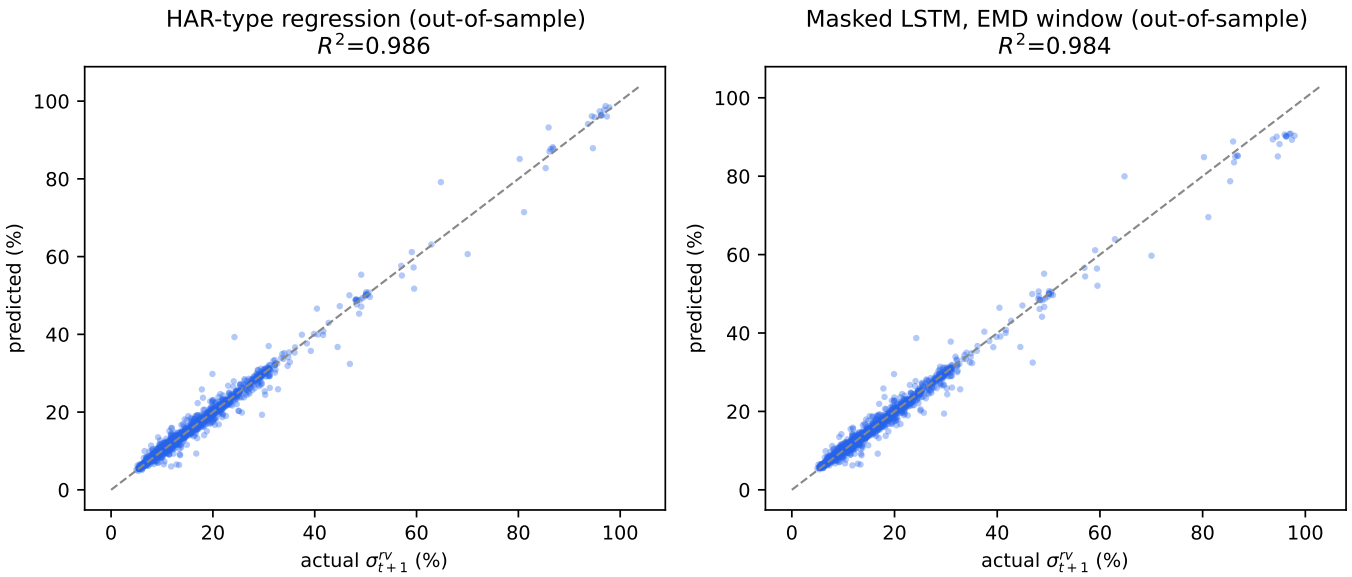


Fig. 2. Out-of-sample predicted vs. realised $\sigma^{\mathrm{rv}}_{t+1}$. Both methods track the diagonal across the full range, including the COVID-period extremes near $\sigma = 100$.

C. Re-estimating every year

Table V repeats stage 1 with every fitted model re-estimated before each of the 9 test years (Section III).

TABLE V

RE-ESTIMATING EVERY MODEL ONCE A YEAR OVER THE TEST PERIOD. STATIC REPEATS TABLES II AND III. YEARLY REFITS GARCH(1,1) AND THE HAR-TYPE REGRESSION ON ALL DATA BEFORE EACH TEST YEAR AND RETRAINS THE NETWORKS ON ALL BUT THE LAST 592 DAYS BEFORE IT, WHICH ARE USED FOR EARLY STOPPING; HYPERPARAMETERS STAY AT THEIR STATIC SELECTION. PARAMETER-FREE ROWS OTHER THAN PERSISTENCE AND THE WINDOW UPDATE ARE UNCHANGED AND OMITTED. p_{Holm} IS AGAINST PERSISTENCE, CORRECTED WITHIN EACH YEARLY TABLE AS IN TABLES II AND III.

	$\sigma^{\mathrm{rv}}_{t+1}$			$\sigma^{\mathrm{rv},(1)}_{t+1}$		
Method	Static	Yearly	p_{Holm}	Static	Yearly	p_{Holm}
Persistence ($\hat{\sigma}_{t+1} = \sigma^{\mathrm{rv}}_t$)	1.795	1.795	–	272.0	272.0	–
Window update (19 of 20 returns known)	0.911	0.911	0.005	–	–	–
GARCH(1,1), QMLE	8.678	10.796	0.013	249.2	247.0	0.269
HAR-type regression on σ^{rv}	1.644	1.631	0.136	263.6	262.6	0.645
Masked LSTM, fixed window	1.861	1.852	1.000	268.6	269.7	1.000
Masked LSTM, EMD dynamic window	1.929	1.902	1.000	266.7	266.0	1.000
LSTM on returns, fixed window	3.188	3.217	0.656	251.1	254.8	0.384

Nothing of substance changes. On the 20-day target, refitting helps the HAR-type regression and the two LSTMs fed σ^{rv} a little (HAR-type 1.631 against 1.644 static), hurts GARCH(1,1) (10.796 against 8.678) and the LSTM on returns, and leaves the window update, which has nothing to refit, far ahead. No fitted model beats persistence after correction. On the single-day target GARCH(1,1) still has the lowest error (247.0) and still does not clear the Holm bar ($p_{\mathrm{Holm}} = 0.269$). The two window rules remain indistinguishable, at $p = 0.11$ on the 20-day target and $p = 0.07$ on the single-day one, where the dynamic window is again slightly ahead but at a different $W_{\max}$ from the fixed one. Holding the models fixed for eight years was not what held the networks back.

D. Stage 2: pricing

Table VI evaluates against V^*_t from (3), which is unknown at t. Because V^* is driven by a single day's return there is an analytic bound on how well anything can do. If $r_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(0, \sigma^2_t)$ then $|r_{t+1}|$ is half-normal with mean $\sigma_t\sqrt{2/\pi}$ and variance $\sigma^2_t(1 - 2/\pi)$, so

$$R^2_{\max} \;=\; \frac{\mathrm{Var}\big(\tfrac{1}{2}S_t\sigma_t\sqrt{2/\pi}\big)}{\mathrm{Var}\big(\tfrac{1}{2}S_t\sigma_t\sqrt{2/\pi}\big) + \mathbb{E}\big[\tfrac{1}{4}S^2_t\sigma^2_t(1-2/\pi)\big]} \;=\; 0.240$$

on this test window, even with a perfect σ_t. Under that assumption no model can exceed it. In practice the value is computed with σ^{rv}_t standing in for the unobservable σ_t and ignores fat tails, so it is an estimate of the ceiling rather than a hard bound.

TABLE VI

ONE-DAY ATM CALL, EX-POST FAIR VALUE $V^*_t = \frac{1}{2}|S_{t+1} - S_t|$, TEST SET OF 622 DAYS. DM IS AGAINST BS (GARCH(1,1)), THE BEST BLACK–SCHOLES VARIANT (NEWEY–WEST, BANDWIDTH AT LEAST 19); POSITIVE MEANS WORSE. NEURAL ENTRIES ARE THE MSE OF THE FIVE-SEED ENSEMBLE FORECAST; $\pm$ GIVES THE S.D. OF THE FIVE SINGLE-SEED MSES. ESTIMATED R^2 CEILING WITH A PERFECT σ_t: 0.240.

Method	MSE	MAE	R^2	DM	p	p_{Holm}
RNN (+LSTM fixed vol)	**377.64 ± 3.18**	13.747	0.140	−2.45	0.014	0.173
RNN (+LSTM dynamic vol)	378.20 ± 2.05	13.744	0.139	−2.46	0.014	0.173
RNN (base)	378.93 ± 3.84	13.798	0.137	−2.48	0.013	0.173
BS (GARCH(1,1))	427.11	14.679	0.028	–	–	–
BS (Yang–Zhang)	434.63	14.351	0.011	+0.70	0.486	1.000
BS (VIX)	445.30	16.426	-0.014	+0.68	0.497	1.000
BS (EWMA)	445.40	14.829	-0.014	+1.68	0.094	0.938
BS (LSTM on returns)	454.66	14.777	-0.035	+1.43	0.154	1.000
BS (window update)	459.35	14.797	-0.046	+1.46	0.143	1.000
BS (LSTM, EMD dynamic)	460.97	14.789	-0.049	+1.43	0.153	1.000
BS (LSTM, fixed window)	461.29	14.793	-0.050	+1.43	0.153	1.000
BS (σ^{rv}_t)	462.23	14.831	-0.052	+1.46	0.145	1.000
BS (HAR-type)	462.41	14.835	-0.053	+1.46	0.145	1.000
Constant (training mean)	463.00	13.536	-0.054	+1.00	0.317	1.000
BS (constant σ)	515.87	17.584	-0.174	+3.63	<0.001	0.004
Persistence (V^*_{t-1})	734.24	18.698	-0.671	+5.02	<0.001	<0.001

BS (constant σ) uses the training-slice mean of σ^{rv}. The constant row is in index points and does not scale with S_t.

Three things stand out. First, conditioning on volatility matters, but the choice of volatility model hardly does. Black–Scholes with a constant σ (the training mean, 18.0%) is clearly worse than Black–Scholes fed with GARCH(1,1) (MSE 515.9 against 427.1, $p_{\mathrm{Holm}} = 0.004$). Among the variants that do receive a volatility estimate, every one lies within 8.3% of the best and

none is significantly different from it. The forecasts that did best on $\sigma^{\mathrm{rv}}_{t+1}$ (the window update, the LSTMs fed σ^{rv}, persistence and the HAR-type regression) sit at the bottom of that group, with GARCH(1,1) and Yang–Zhang at the top. A responsive estimate of tomorrow's variance may simply suit a target driven by one day's return better than a smooth 20-day statistic does, however well that statistic is forecast. None of these differences is significant, so we offer this as a reading, not a finding.

Second, the constant row needs a warning. Predicting the training-mean V^* in index points cannot be told apart from the best Black–Scholes variant ($p = 0.32$), which seems to say that volatility forecasting is worthless here. That is an accident of the test window. The index stands 1.75 times higher than in the training slice while volatility is 22% lower (σ^{rv} averages 13.9% against 18.0%), and in index points the two effects roughly cancel. The constant-σ row scales with S_t and gets no such help, which is why it, and not the constant, is the comparison that isolates the value of a volatility input.

Third, the learned pricer is the best row in the table. All three RNN variants beat every Black–Scholes variant, at an MSE near 378.9 against 427.1 for BS (GARCH), about 11% lower, and an R^2 of 0.137 against 0.028. The raw p-value against BS (GARCH) is 0.013 and the block bootstrap gives 0.037, but after the Holm correction across the table it is 0.173, so this is not a significant result either. It is not a level calibration: rescaling BS (GARCH) by the factor that best fits the training slice (0.993) leaves its MSE at 426.0, and the RNN still beats that (bootstrap $p = 0.040$). What the RNN does not need is the forecast this paper is about. With log return, σ^{rv}_t, VIX and the T-bill yield as inputs it scores 378.9; adding either LSTM forecast gives 377.6 or 378.2, differences much smaller than the seed-to-seed spread of ± 3.8.

V. Discussion: Why the Window Rule Does Not Help

Three diagnostics point to a mechanism.

Figure 3 shows how τ_t actually behaves. With $W_{\max} = 10$, the cap that validation selected on the 20-day target, it sits at the cap 42.3% of the time, with mean 8.27 and standard deviation 1.90. That share says more about the cap than about the rule: $\pi(t)$ has a median of 9.1, so a cap of 10 binds often, while a cap of 20 would bind only 0.8% of the time. The structural problem is elsewhere. $\pi(t)$ in (4) is dimensionless, and because each π_k averages to roughly one, $\pi(t)$ averages to roughly K, the number of IMFs extracted from the trailing window. K grows like $\log_2 L$ in the lookback length L, so the typical window length is set by how much history the decomposition is given, not by any timescale of the series.

$\pi(t)$ does move with the volatility regime, but the wrong way. It correlates at 0.38 with σ^{rv}_t and averages 8.9 on the calmest fifth of days against 11.4 on the most volatile fifth, so turbulent periods get *longer* windows, the opposite of the motivation for the method. With $W_{\max} = 10$ most of that variation is clipped off anyway. That validation preferred the cap that switches off much of the adaptation is a hint in itself.

The third diagnostic is the price of causality. The rule must read $\pi(t)$ at the last sample of its trailing window, where the envelope splines and the FFT-based Hilbert transform are least reliable. For 150 randomly chosen dates we also computed π at the same date from a window reaching 50 days past it (not causal, and used only as a measurement). The two values correlate at 0.51, so roughly half of what the causal rule reacts to is an edge effect. A defensible adaptive rule would target something like the local decorrelation time of the series and would not depend on the end point of a decomposition.

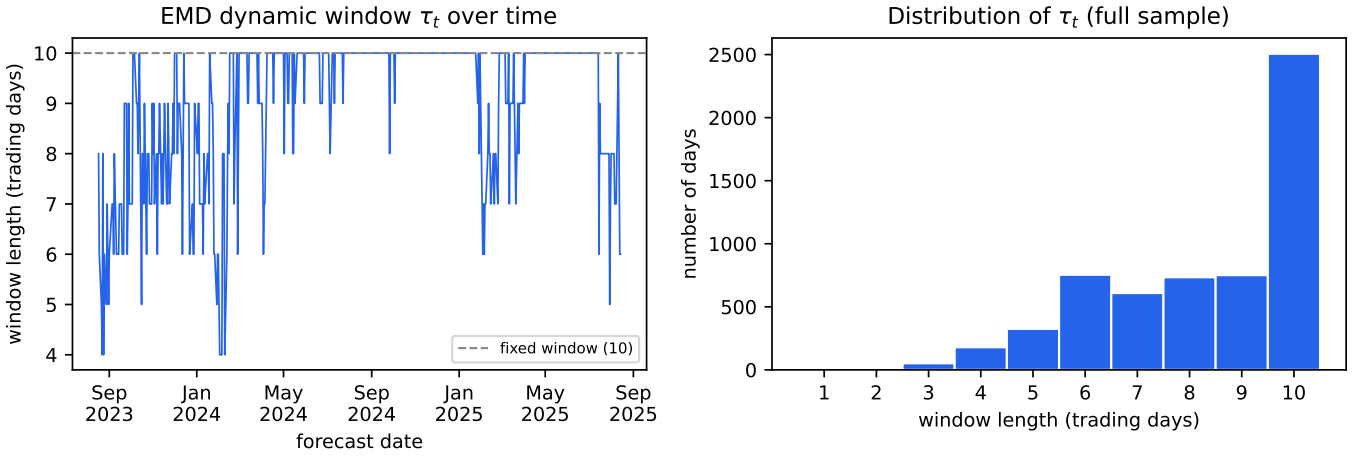


Fig. 3. Left: the causal τ_t over roughly two recent years, against the fixed window it is compared with (both with $W_{\max} = 10$). Right: the full-sample distribution of τ_t, piled up at the cap.

The pricing stage has its own explanation. Figure 4 shows that over the full sample the smoothed volatility estimators and VIX correlate at 0.88 or more, while the single-day proxy, which is mostly noise, reaches at most 0.60. On the pricing test slice the inputs actually fed to Black–Scholes are less alike: still 0.89 or more pairwise without VIX, but as low as 0.61 between VIX and σ^{rv}_t. So collinearity explains part of the flatness of Table VI, not all of it. The larger part is the target. With an R^2

ceiling of about 0.240, a one-day option's ex-post value is mostly one day of noise, and there is little room in which a better volatility forecast could show itself.

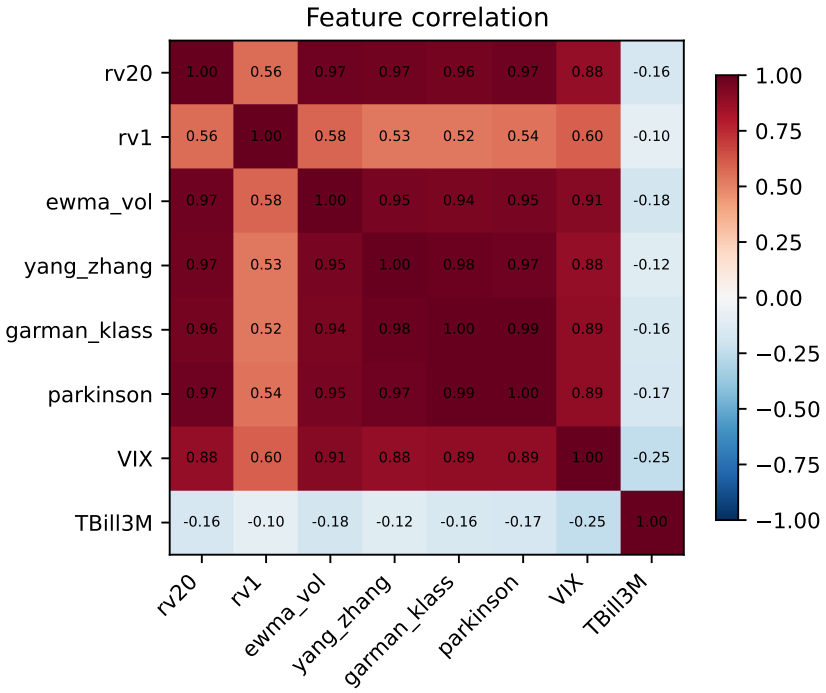


Fig. 4. Correlation among price-derived features over the full sample. The smoothed volatility estimators and VIX cluster tightly; the single-day proxy, being mostly noise, correlates much less; the T-bill yield is mildly anti-correlated with volatility.

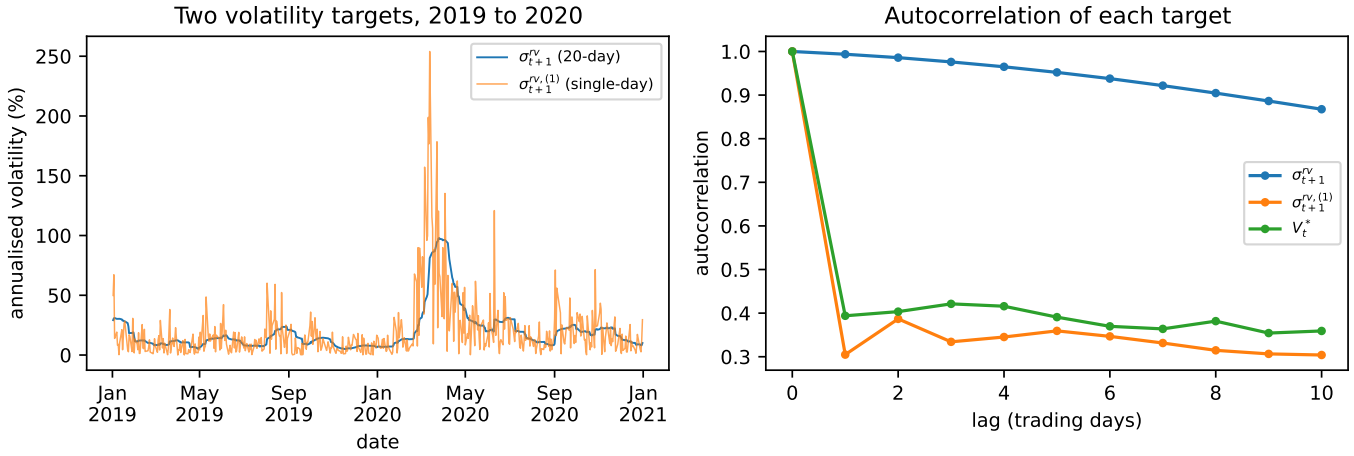


Fig. 5. Left: the two volatility targets over 2019 and 2020 on a common axis. Right: autocorrelation functions. $\sigma^{\mathrm{rv}}_{t+1}$ decays slowly, which is why persistence is so strong on it; $\sigma^{\mathrm{rv},(1)}_{t+1}$ and V^*_t decay within days and are the harder problems.

VI. Design Verification

Because the conclusions rest on the design rather than on the architecture, the design itself is tested. Two sets of checks run before any number in this paper was recorded.

The unit tests (`src/checks/unit_tests.py`) check components against closed forms and simulations with known parameters: the realised-volatility, Parkinson, Garman–Klass and Yang–Zhang estimators each recover the true σ of simulated price paths to within 3% (range estimators are biased low at coarse intraday sampling, a discretisation effect that decays from -8% at 100 steps per day to under -1% at 10,000, so the simulation is refined until the bias is far below tolerance); the QMLE recovers α and β from simulated GARCH(1,1) data; the Black–Scholes routine matches its small-T ATM approximation; the

Diebold–Mariano statistic is zero for identical error series and significant for a genuinely worse one; in simulation with a persistent loss differential and a true null, the test without HAC correction rejects 31% of the time at the 5% level, while the 19-lag floor brings that to 7%; the block bootstrap holds its size under the null and detects a model with twice the error; the window update reproduces $\sigma^{\mathrm{rv}}_{t+1}$ exactly when the next return is zero and never reads past t; the QLIKE variance proxy on the single-day target is unbiased; and the masking residual described in Section II is measured at 4×10^{-3}.

The audit (`src/checks/audit.py`) checks the assembled study: that V_t^* equals $\frac{1}{2}|S_{t+1} - S_t|$ exactly and σ^{rv} targets are aligned to $t+1$ with no off-by-one; that no target is an exact algebraic identity of same-row features; that $\pi(t)$ is unchanged when the series is corrupted after t; that windows, including the two-feature windows of the returns LSTM, never reach past the forecast origin; that the stage-1 test forecasts start where validation ends, the pricing test period lies inside them, and the yearly-refit forecasts cover exactly the same days; that the input scaler is unchanged when every value after the training slice is corrupted, and that the target scaler stored in the network equals the training-target mean; that every results table contains a persistence benchmark and corrected p-values, the 20-day table the window update, and the pricing table both naive and no-forecast benchmarks; that every neural configuration carries five seeds; that every comparison uses the same Newey–West floor; and that every volatility column shares one set of units. All checks pass.

The tables in this paper are generated directly from the result files, and the quantities quoted in the text are LaTeX macros emitted by the same step, so no number here is transcribed by hand.

VII. Limitations

- **No traded option data.** We never touch a traded option price. The pricing target is an ex-post construction from the underlying, so we can say nothing about bid–ask spreads, the volatility smile, or whether any of this survives execution. A stronger version of this study needs OptionMetrics or an equivalent panel.
- **The window rule has no derivation.** As Section II notes and Figure 3 confirms, $\pi(t)$ has no units of time and its magnitude is governed by the EMD lookback length. Our negative result is a result about *this* rule, not about adaptive windows in general.
- **The validation slice is calmer than the test period.** With a chronological split the validation window reaches a maximum σ^{rv} of about 27 while the test period reaches nearly 100, so early stopping is tuned on a quieter regime than the one it is evaluated on. This is unavoidable when the most volatile episode in the sample lies in the test period, but it is a real caveat on every neural row.
- **How re-estimation was done.** The yearly refit in Section IV-C keeps hyperparameters at their static selection, and the networks give up their most recent 592 days to early stopping while GARCH and the HAR-type regression use all data up to each refit date. Re-running the grid search every year, or retraining on all data for a fixed number of epochs, could treat the networks more generously. We doubt it would change the picture, since refitting changed none of the conclusions.
- **How the returns enter the network.** The LSTM on returns receives standardised absolute returns, and the 2020 crash pushed them far outside the training range. A log or rank transform, or clipping, would probably tame that episode. It would not explain the gap to the window update in calmer periods.
- **Single instrument, single horizon.** One index and a one-day horizon. One day is also the horizon at which tomorrow's volatility is closest to today's, which is why the fitted models struggle to separate from persistence. A 10- or 20-day forward target would make the forecasting problem more interesting, at the cost of overlapping targets that need block-bootstrap inference.
- **Point forecasts only.** Stage 1 estimates a scale parameter, so the natural output is a predictive distribution scored by CRPS or a quantile loss rather than a point forecast scored by MSE. That would also sidestep the fact that MSE on V^* is dominated by irreducible noise.
- **Risk-free rate convention.** We use the 13-week T-bill discount yield directly as a continuously compounded rate, without converting between quoting conventions. At a one-day horizon the effect is far below the differences under discussion, but it is a simplification.
- **Architecture.** Attention-based sequence models are the obvious comparison and are not tested here. On roughly 5,921 daily observations we expect sample size rather than architecture to bind. Studies that do find neural gains tend to use intraday data or many assets at once [16], [9], which is where we would look next.

VIII. Reproducibility

The whole study runs from one command, `python run.py all`, which executes in order

`features` → `pi` → `stage1` → `stage2` → `refit` → `robustness` → `test` → `audit` → `tables` → `figures` → `paper`

and stops if a unit test or audit check fails. The data download (`python run.py fetch`) is kept separate because it depends on the date it is run. Every setting (paths, seeds, splits, grids, bandwidths) lives in `src/config.py`. Every neural

forecast is a seeded five-network ensemble, and starting again from the raw data reproduces every number in this paper exactly, given the same software versions. Per-seed results, the selected hyperparameters and every test-set prediction are saved under `results/`, so any table can be re-derived without retraining. Re-running the data fetch on a later date pulls additional trading days and shifts point estimates slightly. The conclusions that matter here, namely which comparisons survive correction and the absence of any advantage for the dynamic window, were stable across re-runs during development.

IX. Conclusion

We set out to test whether an EMD-driven adaptive sliding window improves a masked LSTM's volatility forecast enough to produce a better option price. It does not. Tested directly against a fixed window of the same architecture, the adaptive window is indistinguishable on both volatility targets, barely changes the forecast, and the sign of the difference is not even consistent between them. Neither variant separates from persistence, and on the 20-day target both are far behind a parameter-free update of the rolling window. A network given the daily returns does not catch up with that update either, and breaks down in the 2020 crash. On the single-day target GARCH(1,1) does best, with the returns-fed network close behind, neither clearing the multiple-comparison bar. Re-estimating every model each year changes none of these conclusions. At the pricing stage, conditioning on volatility clearly helps but the choice among volatility models does not, and a small recurrent pricer that has no use for our forecast does best of all, again without significance after correction.

The diagnostics suggest why. The window rule's typical length is set by the number of extracted modes, which depends on the lookback length rather than the market; what regime dependence it has points the wrong way; and about half of its movement at the end of a causal window is an edge effect. At the pricing stage the target is mostly noise, with an R^2 ceiling of about 0.240, which leaves little room for a better forecast to matter. The result is bounded: one index, one horizon, one window rule. It says that this rule does not work and why, not that adaptive windows cannot work. What it does say more generally is that on a rolling-window target the naive benchmark has to use what is already known about tomorrow's window, that the bandwidth of a Diebold–Mariano test can decide its verdict, and that a correction for the number of comparisons is what separates a result from an artefact.

Acknowledgment

Thanks to Professor Liao for her guidance throughout this project.

All code is openly available at https://github.com/PaulAnrd/Research_Project.

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